# SEPARATION OF A K-RATIONAL POINT FROM FINITELY MANY POINTS BY A SECTION OF A VERY AMPLE INVERTIBLE SHEAF

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The author disposes of the case n = 0 with the single word 'trivial' and proceeds; that sentence is reproduced as written and no case is dropped.

*Editorial note.* The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble: Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's no-op display environments for slogan/reference/history.

*Editorial note.* The label lemma-very-ample-vanish-at-point is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

*Allowed standard prerequisites (editorial).*

- Very ample invertible sheaf on a finite type k-scheme. Same chapter, the definition of a very ample invertible sheaf as one whose sections give a closed immersion into projective space. Supplies the reduction of the statement to the case X = P^r_k with L = O_X(1).
- Homogeneous coordinate ring and Proj of P^r_k. Same chapter, P^r_k = Proj(k[T_0, ..., T_r]) and the correspondence between points and homogeneous prime ideals. Supplies the identification of the sections vanishing at x with a codimension one subspace W of V = kT_0 + ... + kT_r.
- A finite dimensional vector space over an infinite field is not a finite union of proper subspaces. Standard linear algebra, used without a pointer at proof lines 4255-4256. Supplies the conclusion that W is not the union of the finitely many W cap p_i.

**Lemma 0.1.** *Let $k$ be an infinite field. Let $X$ be a scheme of finite type over $k$. Let $\mathcal{L}$ be a very ample invertible sheaf on $X$. Let $n \geq 0$ and $x, x_1, \ldots, x_n \in X$ be points with $x$ a $k$-rational point, i.e., $\kappa(x) = k$, and $x \neq x_i$ for $i = 1, \ldots, n$. Then there exists an $s \in H^0(X, \mathcal{L})$ which vanishes at $x$ but not at $x_i$.*

*Proof.* If $n = 0$ the result is trivial, hence we assume $n > 0$. By definition of a very ample invertible sheaf, the lemma immediately reduces to the case where $X = \mathbf{P}^r_k$ for some $r > 0$ and $\mathcal{L} = \mathcal{O}_X(1)$. Write $\mathbf{P}^r_k = \text{Proj}(k[T_0, \ldots, T_r])$. Set $V = H^0(X, \mathcal{L}) = kT_0 \oplus \ldots \oplus kT_r$. Since $x$ is a $k$-rational point, we see that the set $s \in V$ which vanish at $x$ is a codimension 1 subspace $W \subset V$ and that $W$ generates the homogeneous prime ideal corresponding to $x$. Since $x_i \neq x$ the corresponding homogeneous prime $\mathfrak{p}_i \subset k[T_0, \ldots, T_r]$ does not contain $W$. Since $k$ is infinite, we then see that $W \neq \bigcup W \cap \mathfrak{q}_i$ and the proof is complete. □

*Date*: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.